

HTML CSS & JS

Introduction to HTML

What is HTML?

- HTML stands for HyperText Markup Language.
- It is the standard language for creating web pages.
- HTML describes the structure of a web page using elements and tags.
- It is not a ~~programming language~~ rather a **markup language**.

Why HTML

- Important for front-end web development
- Works together with
 - CSS – styling
 - JavaScript – Interactivity

Role of HTML in Web Development

- HTML – Structure and content
- CSS – Styling and layout of the content
- JavaScript – Interactivity and dynamic behavior

```
1  <!DOCTYPE html>
2  <html>
3    <head>
4      <title>My First Web Page</title>
5      <link rel="stylesheet" href="styles.css">
6    </head>
7    <body>
8      <h1>Welcome to My Website</h1>
9      <p>This is a paragraph.</p>
10     <script src="script.js"></script>
11   </body>
12 </html>
```

Basic Structure of an HTML Document

- `<!DOCTYPE html>` - Declares the document type and version
 - Tells the browser which version of HTML the page is using
- `<html>` - The root element of the document
- `<head>` - Contains metadata (simply data that describes data)
 - E.g. title, stylesheets, scripts
- `<body>` - Contains the visible content of the web page

The !DOCTYPE

- Different versions of HTML have been designed since the creation of the web
 - Each new version was an improvement on the last version
- Read more on the evolution of HTML for more information
 - (Duckett, 2011, p. 199)
- Refer on the figure on how each HTML version is specified using !DOCTYPE

HTML5	HTML
<!DOCTYPE html>	
HTML 4	
<!DOCTYPE html PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN" "http://www.w3.org/TR/html4/loose.dtd">	
Transitional XHTML 1.0	
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/ xhtml1-transitional.dtd">	
Strict XHTML 1.0	
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN" "http://www.w3.org/TR/xhtml1/DTD/ xhtml1-strict.dtd">	
XML Declaration	
<?xml version="1.0" ?>	

The `<head>` Section

- Its content is not visible in the web browser when the page is loaded.
- Contains metadata about the HTML (e.g. author, and keywords that describe the document, character set, viewport settings), links to CSS and favicons, and scripts.
- Common Tags in `<head>`
 - `<head>` - Sets the title of the page (shown in the browser tab)
 - `<meta>` - Provides metadata
 - `<link>` - Links external resources like CSS files
 - `<script>` - Links JavaScript files

The <head> Section

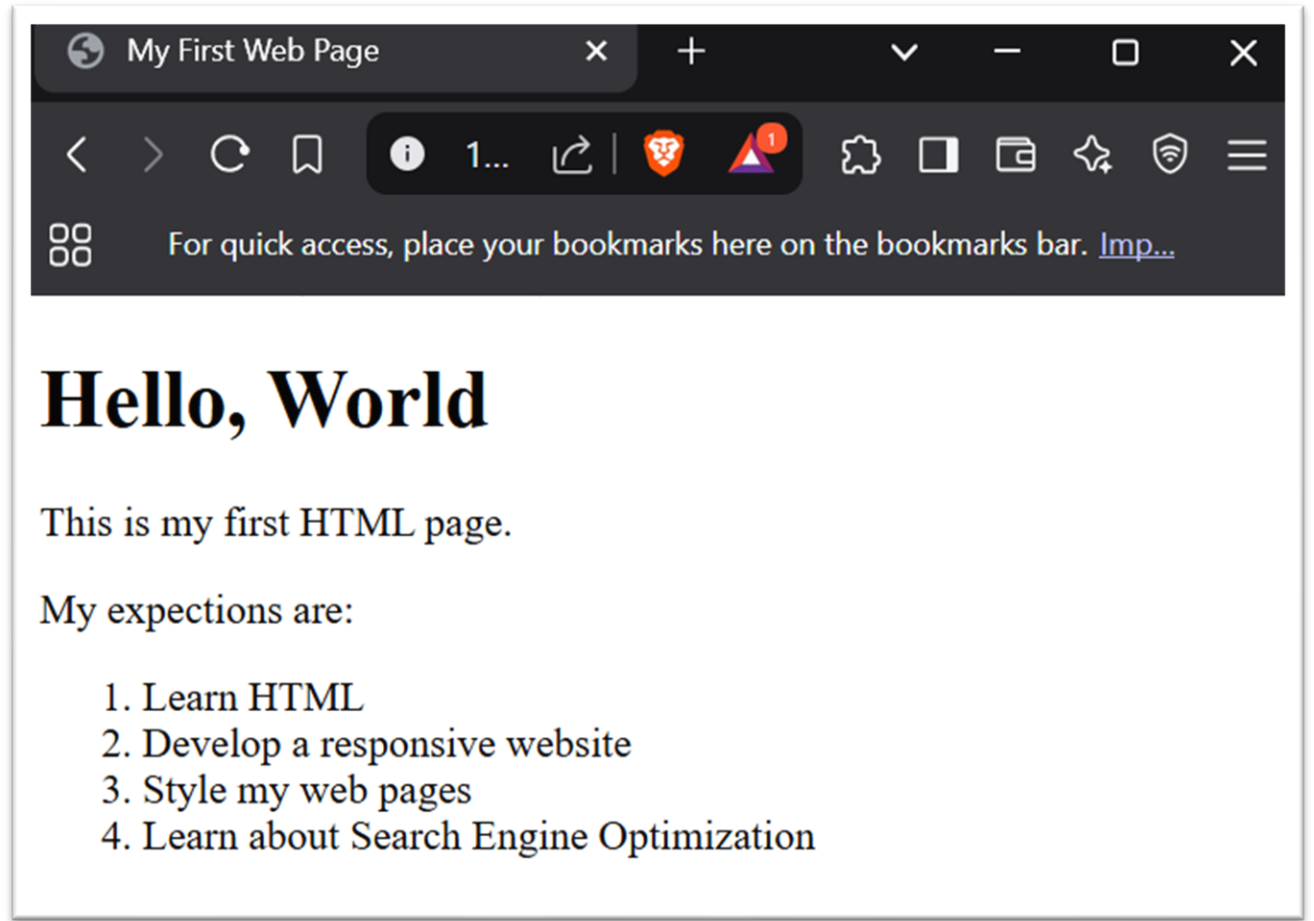
```
1  <!DOCTYPE html>
2  <html>
3    <head>
4      <title>My First Web Page</title>
5      <meta charset="UTF-8">
6      <meta name="viewport" content="width=device-width, initial-scale=1.0">
7      <link rel="stylesheet" href="styles.css">
8      <script src="script.js"></script>
9    </head>
10 > <body> ...
14 </body>
15 </html>
```

The `<body>` Section

- Contains the visible content of the web page
- There can only be one `<body>` element in a document
- It has content sectioning elements.
- Common elements in `<body>`
 - `<h1>` to `<h6>` - Headings
 - `<p>` - Paragraphs
 - `<a>` - Links
 - `<img>` - Images
 - `<ul>`, `<ol>`, `<li>` - Lists
 - `<nav>`, `<header>`, `<footer>` - Sections

The `<body>` Section

```
1 <!DOCTYPE html>
2 <html>
3 > <head> ...
9 </head>
10 <body>
11 <h1>Hello, World</h1>
12 <p>This is my first HTML page.</p>
13 <p>My expectations are: </p>
14 <ol>
15 <li>Learn HTML</li>
16 <li>Develop a responsive website</li>
17 <li>Style my web pages</li>
18 <li>Learn about Search Engine Optimization</li>
19 </ol>
20 </body>
21 </html>
```

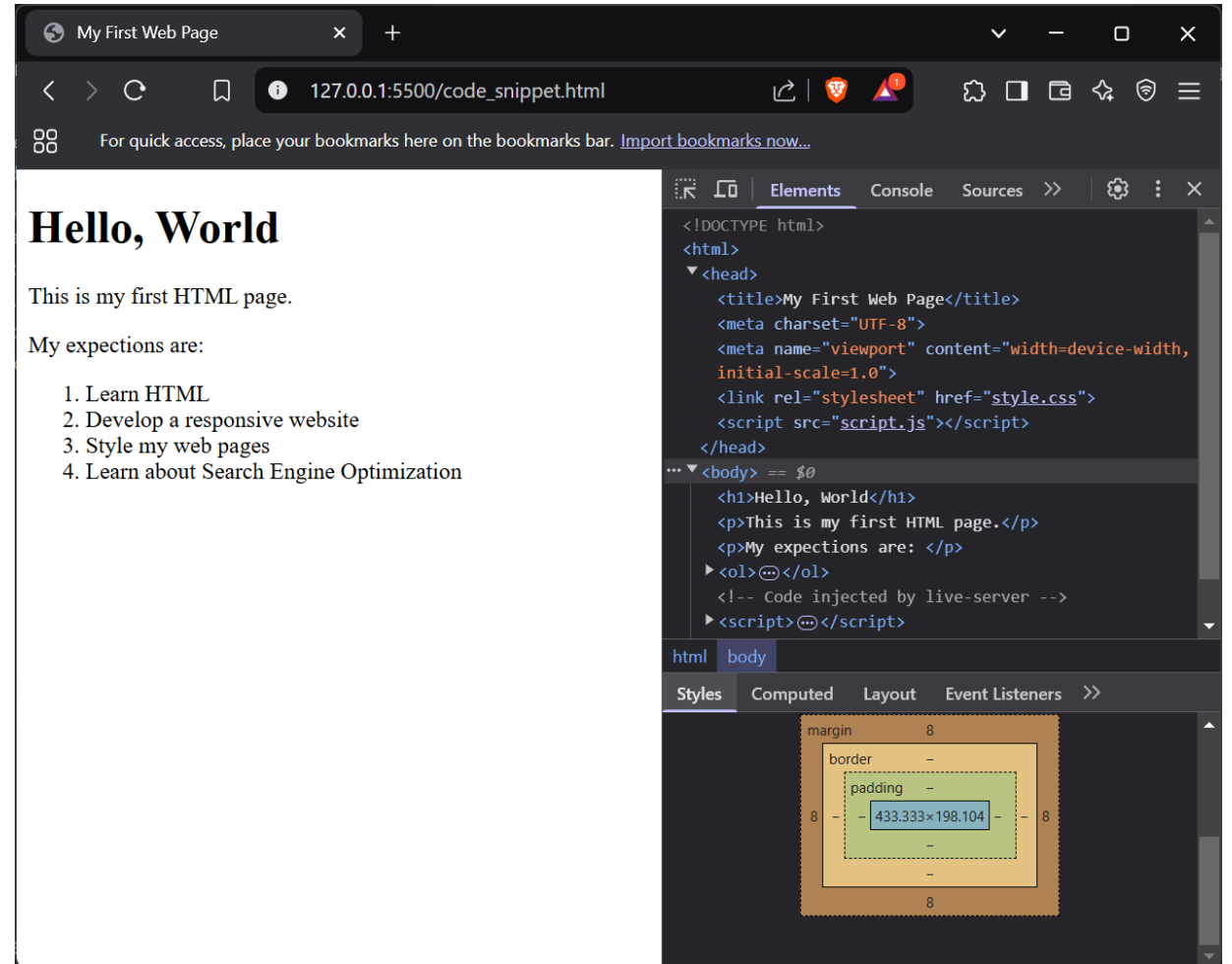


HTML Tools

- Text Editors
 - VS Code – highly recommended
 - Sublime Text – Lightweight and fast
 - Atom – Open-source and customizable
- Browser Developer Tools
 - Inspect Element – View and edit HTML/CSS in real-time
 - Console – Debug JavaScript errors

HTML Tools

- Browser Developer Tools
 - Inspector: - right click any blank place and then click **inspect**



Caveat

- Putting elements in a wrong section will raise errors
- However, modern browsers correct that by moving the elements to their right section, or ? (*read more and observe by trying*)
 - Try putting a `<h1>` p in the element, then after running the server, inspect the page.
 - Observe the difference between your HTML document and the HTML shown during inspection.
 - Also try moving from the `<head>` section to `<body>` section
 - Observe what is happening

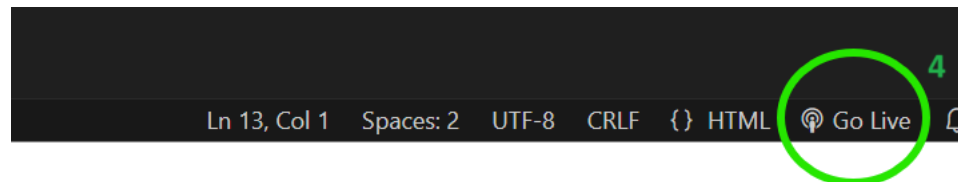
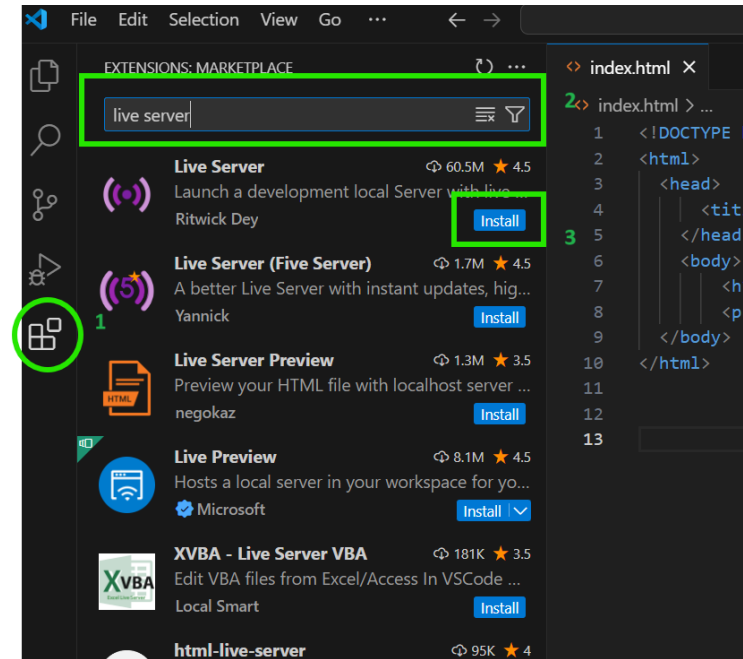
Creating Your HTML File

- Open a text editor, for instance VS Code
- Create a new file and save it as ***index.html***
- Write the basic HTML structure as shown to your write

```
<> index.html > ...
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>My First Web Page</title>
5  </head>
6  <body>
7      <h1>Hello, World</h1>
8      <p>This is my first HTML page.</p>
9  </body>
10 </html>
11
```

Running Your HTML file

- Running on localhost server
 1. Add a **live server** extension in VS Code
 2. Click on the **Go Live** button at the right bottom of the VS Code
 3. Then a browser will run the render the HTML document



References

- <https://developer.mozilla.org/en-US/docs/Web/HTML/Element>
- Duckett, J. (2011). *HTML & CSS - Design and Build Websites*. Indianapolis, USA: John Wiley & Sons, Inc.